

AI FAQ ASSISTANT API

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PHASE 1 – BRAINSTORMING & IDEATION

Project Title

AI FAQ Assistant

Problem Statement

Customers and users often have to search through large amounts of information to find answers to frequently asked questions. Traditional FAQ systems mainly provide predefined answers and may not understand questions asked in different ways.

Proposed Solution

The AI FAQ Assistant is an AI-powered REST API application that provides intelligent FAQ and customer-support functionality. It allows users to manage FAQs and uses AI to generate relevant responses to user queries.

Objectives

- To develop an AI-powered FAQ and customer support system.
- To provide quick and relevant answers to user queries.
- To reduce repetitive customer-support work.
- To securely manage users using authentication.
- To store and manage FAQ information using MongoDB.
- To provide REST APIs for easy integration with other applications.

Main Features

- User registration and login.
- JWT-based authentication.
- FAQ creation and management.
- AI-powered question answering.
- MongoDB database integration.
- RESTful API architecture.
- Secure API access.
- Error handling and validation.

Expected Outcome

The final system should provide a secure and intelligent backend API that can manage FAQs and assist users by generating relevant AI-based responses.

PHASE 2 – REQUIREMENT ANALYSIS

Functional Requirements

1. User Registration

- New users should be able to create an account.
- User details should be stored securely in the database.

2. User Login

- Registered users should be able to log in.
- The system should authenticate users and generate a JWT token.

3. Authentication

- Protected APIs should be accessible only to authenticated users.
- JWT should be used for secure authentication.

4. FAQ Management

- Users should be able to create, view, update, and delete FAQ information based on the implemented access rules.

5. AI Question Answering

- The system should accept user questions.
- The Gemini API should process the request and generate an appropriate response.

6. Database Management

- User and FAQ information should be stored in MongoDB.

7. Error Handling

- The application should provide meaningful error messages for invalid requests and server errors.

Non-Functional Requirements:

- Security
- Reliability
- Maintainability
- Scalability
- Performance
- Usability

Software Requirements:

1. Node.js
2. Express.js
3. MongoDB
4. Mongoose
5. JavaScript
6. Gemini API
7. JWT
8. Postman
9. Visual Studio Code
10. Git and GitHub

Hardware Requirements:

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- Sufficient storage for project files and dependencies

PHASE 3 – PROJECT DESIGN

System Architecture

The AI FAQ Assistant follows a backend REST API architecture.

Architecture Flow

User/Client



REST API



Express.js Server



Authentication & API Logic



MongoDB Database / Gemini API



Response to User

Main Modules

1. Authentication Module

Handles user registration, login, password security, and JWT authentication.

2. FAQ Module

Handles creation, retrieval, updating, and deletion of FAQ information.

3. AI Module

Receives user questions and communicates with the Gemini API to generate AI-based responses.

4. Database Module

Uses MongoDB with Mongoose to store and manage application data.

5. Error Handling Module

Handles validation errors, authentication errors, API errors, and server errors.

Database Design

The application uses MongoDB as the primary database.

User Collection

- User ID
- Name
- Email
- Password
- Created Date

FAQ Collection

- FAQ ID
- Question
- Answer
- Category
- Created Date
- Updated Date

API Design

The application exposes RESTful endpoints for authentication, FAQ management, and AI-powered question answering.

PHASE 4 – PROJECT PLANNING

Project Development Plan

Phase - Activities

1. Ideation - Identify problem and propose AI FAQ solution
2. Requirement Analysis- Identify functional and technical requirements
3. Design - Design architecture, database and API flow
4. Planning - Divide project into modules and development tasks
5. Development - Implement backend APIs and AI integration
6. Testing - Test APIs and fix errors
7. Documentation - Prepare project report and supporting documents
8. Demonstration -Demonstrate the completed project

Development Modules:

1. Project setup
2. Express server configuration
3. MongoDB connection
4. User authentication
5. JWT authorization
6. FAQ APIs
7. Gemini API integration
8. Error handling
9. API testing
10. Documentation

Tools and Technologies:

- Frontend/Client: API client such as thunder client
- Backend: Node.js and Express.js
- Database: MongoDB
- ODM: Mongoose
- Authentication: JWT
- AI Service: Gemini API

- Development Tool: Visual Studio Code
- Version Control: GitHub

Expected Deliverables:

- Source code
- Database implementation
- API endpoints
- Test results
- Project documentation
- Public GitHub repository
- Project demonstration video

PHASE 5 – PROJECT DEVELOPMENT

Development Overview

The AI FAQ Assistant was developed as a REST API application using Node.js and Express.js. MongoDB was used as the database and Mongoose was used for database interaction.

Development Steps

Step 1 – Project Setup

The Node.js project was initialized and the required dependencies were installed.

Step 2 – Express Server

An Express.js server was created to handle HTTP requests and API routes.

Step 3 – MongoDB Integration

MongoDB was connected to the application using Mongoose.

Step 4 – User Authentication

Registration and login functionality were implemented. JWT was used to authenticate users and protect authorized routes.

Step 5 – FAQ Management

API endpoints were developed to manage FAQ data.

Step 6 – Gemini AI Integration

The Gemini API was integrated to process user questions and generate AI-based responses.

Step 7 – Error Handling

Validation and error-handling mechanisms were implemented to handle invalid requests and application errors.

Step 8 – API Testing

The developed APIs were tested using Postman to verify their functionality.

Development Outcome

The backend application successfully provides authentication, FAQ management, database connectivity, and AI-powered question answering through REST APIs.

PHASE 6 – PROJECT TESTING

Testing Objective

Testing was performed to verify that all major functions of the AI FAQ Assistant work correctly and that the APIs return the expected responses.

Testing Method

The REST APIs were tested using Postman.

Test Cases

Test Case - Expected Result

User Registration - User should be registered successfully

User Login - Valid credentials should return authentication token

Invalid Login - Invalid credentials should return an error

Protected API without Token - Request should be rejected

Create FAQ - FAQ should be created successfully

Get FAQ - FAQ information should be retrieved

Update FAQ - Existing FAQ should be updated

Delete FAQ - FAQ should be deleted according to access rules

AI Question - AI-generated response should be returned

Invalid Request - Appropriate validation/error message should be displayed

Testing Result

The APIs were tested with valid and invalid inputs. Authentication, FAQ operations, database operations, and AI response functionality were verified. Errors identified during testing were corrected during development.

Evidence

The following screenshots can be attached:

- Registration API response
- Login API response
- JWT token response
- Create FAQ response
- Get FAQ response
- AI question/answer response
- Error response

Project Overview

The AI FAQ Assistant is an AI-powered REST API designed to provide FAQ management and intelligent customer-support functionality.

Technologies Used:

- Node.js
- Express.js
- MongoDB
- Mongoose
- JavaScript
- JWT
- Gemini API
- Thunder client
- GitHub

Project Objectives

The main objective is to develop a secure and intelligent backend system that can manage FAQs and provide AI-powered responses to user queries.

System Modules

1. User Authentication
2. JWT Authorization
3. FAQ Management
4. AI Question Answering
5. Database Management

6. Error Handling

Project Workflow

User Request



Express.js API



Authentication / Request Processing



MongoDB or Gemini API



Response



User

Advantages:

- Provides quick responses to user queries.
- Reduces repetitive support work.
- Provides centralized FAQ management.
- Uses authentication for protected APIs.
- Can be integrated with other applications through REST APIs.
- AI integration makes the system more flexible than a simple predefined FAQ system.

Future Enhancements:

- Develop a web or mobile frontend.

- Add an admin dashboard.
- Add analytics for frequently asked questions.
- Support multiple languages.
- Add conversation history.
- Improve role-based access control.
- Add more advanced AI-based customer-support features.

Conclusion:

The AI FAQ Assistant demonstrates how REST API development, database management, authentication, and generative AI can be combined to build an intelligent customer-support solution.

PHASE 8 – PROJECT DEMONSTRATION

Demo Video Voice-Over

My project title is AI FAQ Assistant.

The main purpose of this project is to provide an AI-powered FAQ and customer-support solution through REST APIs.

This project is developed using Node.js and Express.js for the backend, MongoDB for database management, Mongoose for database interaction, JWT for authentication, and the Gemini API for AI-powered responses.

First, I will demonstrate the user authentication process.

Here, a user can register with the required details. After successful registration, the user can log in using valid credentials. The application generates a JWT token for authenticated access.

Next, I will demonstrate the FAQ management functionality. The application provides APIs to create, retrieve, update, and delete FAQ information based on the implemented access rules.

Now, I will demonstrate the AI functionality. When a user sends a question, the application processes the request and communicates with the Gemini API. The generated response is then returned to the user.

I have also tested the APIs using Postman with both valid and invalid inputs to verify the functionality and error handling.

The main benefits of this project are faster responses, centralized FAQ management, reduced repetitive customer-support work, and AI-powered question answering.

Finally, this is the working output of my AI FAQ Assistant project.

DEMO VIDEO LINK

<https://drive.google.com/file/d/1wa-JngFz1-us8XxsIgpQMg3Ov3iKswj9/view?usp=drivesdk>

Thank you.